

National University of Computer and Emerging Sciences



Peshawar Campus

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Section: BSSE-223A

Subject: PF Lab Assignment 02

Problem: 1 | Cricket Match

```
#include<stdio.h>
```

```
int main()
```

```
{
```

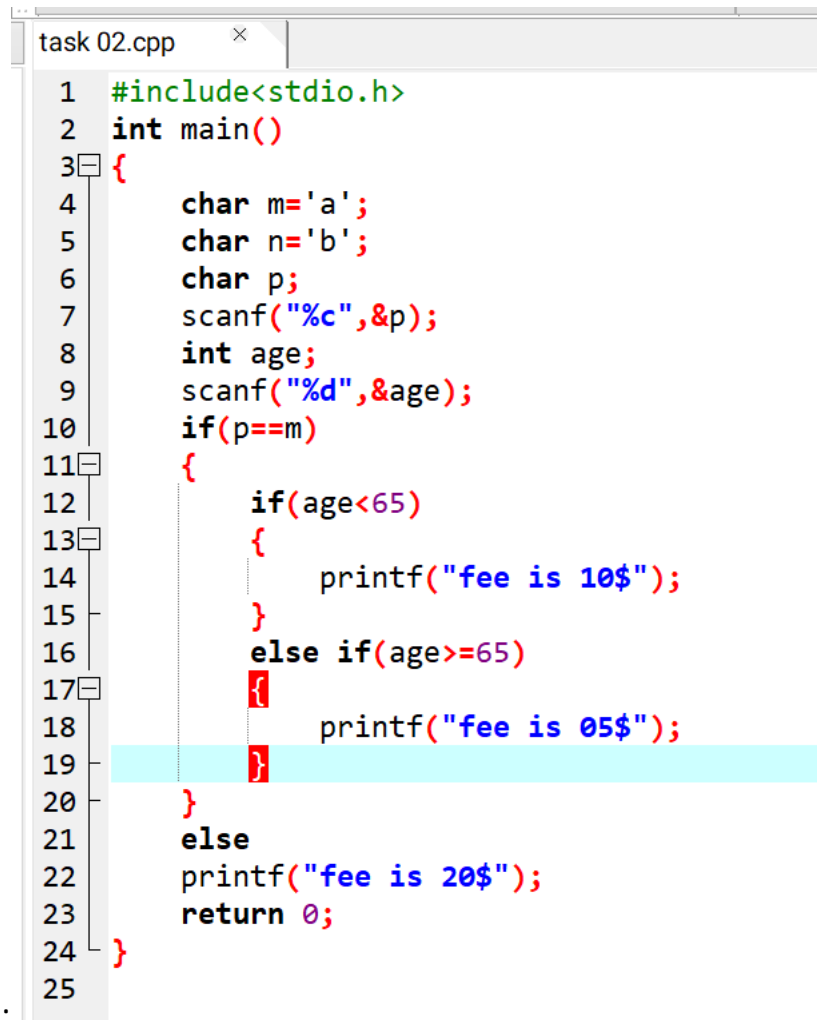
```
int s;
```

```
scanf("%d",&s);
```

```
int w;
```

```
scanf("%d",&w);
```

```
int s_req=300-s;
```

A screenshot of a code editor window titled 'task 02.cpp'. The editor displays a C++ program with line numbers 1 through 25 on the left margin. The code is as follows:

```
1  #include<stdio.h>
2  int main()
3  {
4      char m='a';
5      char n='b';
6      char p;
7      scanf("%c",&p);
8      int age;
9      scanf("%d",&age);
10     if(p==m)
11     {
12         if(age<65)
13         {
14             printf("fee is 10$");
15         }
16         else if(age>=65)
17         {
18             printf("fee is 05$");
19         }
20     }
21     else
22     printf("fee is 20$");
23     return 0;
24 }
25
```

The code is color-coded: keywords like 'include', 'int', 'if', 'else', 'return', and 'scanf' are in blue; variables and literals are in black; and string literals are in red. The closing brace of the main function on line 24 is highlighted with a light blue background.

```
if(s>=0&&w>=0)
{
    if(s>=300&&w>0)
    {
        printf("pakistan won by %d wickets",w);
    }
    else if(s=300&&w==0)
    {
        printf("match is draw");
    }
    else if(s<300&&s>=0)
    {
        printf("pakistan need %d runs to win",s_req);
    }
    }
    else
    printf("wrong input");
    return 0;
}
```

Problem: 2 | (if with logical operators)

```
#include<stdio.h>

int main()
{
    char m='a';
    char n='b';
    char p;
    scanf("%c",&p);
    int age;
    scanf("%d",&age);
    if(p==m)
    {
        if(age<65)
        {
            printf("fee is 10$");
        }
        else if(age>=65)
        {
            printf("fee is 05$");
        }
    }
    else
    printf("fee is 20$");
    return 0;
}
```

```
task 01.cpp
1  #include<stdio.h>
2  int main()
3  {
4      int s;
5      scanf("%d",&s);
6      int w;
7      scanf("%d",&w);
8      int s_req=300-s;
9      if(s>=0&&w>=0)
10     {
11         if(s>=300&&w>0)
12         {
13             printf("pakistan won by %d wickets",w);
14         }
15         else if(s=300&&w==0)
16         {
17             printf("match is draw");
18         }
19         else if(s<300&&s>=0)
20         {
21             printf("pakistan need %d runs to win",s_req);
22         }
23     }
24     else
25     printf("wrong input");
26     return 0;
27 }
```

Problem: 3 | Quadratic Equation

```
#include<stdio.h>

#include<math.h>

int main(){

    int i = -1;

    float root1, root2;

    float a, b, c;

    int discriminant;

    printf("Enter Values : ");
```

```

scanf("%f",&a);
scanf("%f",&b);
scanf("%f",&c);

int Discriminant_value = (b*b)-(4*a*c);

if(a==0 || b==0 || c==0){
printf("Values can't be zero");}

else{

if(Discriminant_value>0){
root1 = ((-b+(sqrt(Discriminant_value)*i))/(2*a));
root2 = ((-b-(sqrt(Discriminant_value)*i))/(2*a));
printf("\nDisc : %d\nRoot 1 : %.2f\nRoot 2 : %.2f",Discriminant_value,root1,root2); }

if(Discriminant_value==0){
root1 = -b/(2*a);
root2 = root1;
printf("Root 1 : %.2f\nRoot 2 : %.2f",root1,root2);}

if(Discriminant_value<0){
root1 = ((-b+sqrt(Discriminant_value)))/(2*a);
root2 = (-b-sqrt(Discriminant_value))/(2*a);
printf("\nDisc : %d\nRoot 1 : %.2fi\nRoot 2 : %.2fi",Discriminant_value,root1,root2);}

}

return 0;

}

```

```

task 03.cpp
1  #include<stdio.h>
2  #include<math.h>
3  int main(){
4      int i = -1;
5      float root1, root2;
6      float a, b, c;
7      int discriminant;
8      printf("Enter Values : ");
9      scanf("%f",&a);
10     scanf("%f",&b);
11     scanf("%f",&c);
12     int Discriminant_value = (b*b)-(4*a*c);
13     if(a==0 || b==0 || c==0){
14         printf("Values can't be zero");}
15     else{
16         if(Discriminant_value>0){
17             root1 = ((-b+(sqrt(Discriminant_value)*i))/(2*a));
18             root2 = ((-b-(sqrt(Discriminant_value)*i))/(2*a));
19             printf("\nDisc : %d\nRoot 1 : %.2f\nRoot 2 : %.2f",Discriminant_value,root1,root2); }
20         if(Discriminant_value==0){
21             root1 = -b/(2*a);
22             root2 = root1;
23             printf("Root 1 : %.2lf\nRoot 2 : %.2lf",root1,root2);}
24         if(Discriminant_value<0){
25             root1 = ((-b+sqrt(Discriminant_value))/(2*a));
26             root2 = ((-b-sqrt(Discriminant_value))/(2*a));
27             printf("\nDisc : %d\nRoot 1 : %.2fi\nRoot 2 : %.2fi",Discriminant_value,root1,root2);}
28     }
29     return 0;
30 }
31

```

Problem: 4 | Largest palindrome product

```

#include<stdio.h>

int count(int num);

int rev_dig(int num,int p);

void printmax();

int main()
{

```

```
printmax();
```

```
}
```

```
int count(int num)
```

```
{
```

```
    int count=0;
```

```
    while(num!=0)
```

```
    {
```

```
        num=num/10;
```

```
        count++;
```

```
    }
```

```
    return (count-1);
```

```
}
```

```
int rev_dig(int num, int p)
```

```
{
```

```
    int res;
```

```
    int sum=0;
```

```
    for(int i=p;i>=0;i--)
```

```
    {  
        res=(num%10);  
        for(int j=i;j>=1;j--)  
            res=res*10;  
        sum=sum+res;  
        num=num/10;  
    }  
    return sum;  
}
```

```
void printmax()  
{  
    int prod;  
    int max=0;  
    int a;  
    int m;  
    for(int i=100;i<=999;i++)  
    {  
        for(int j=100;j<=999;j++)  
        {  
            prod=i*j;  
            a=count(prod);
```

```

        m=rev_dig(prod,a);
        if(m==prod)
        {
            if(m>max)
            max=m;
        }
    }
    printf("%d",max);
}

```

task 04.cpp

```

1  #include<stdio.h>
2  int count(int num);
3  int rev_dig(int num,int p);
4  void printmax();
5  int main()
6  {
7      printmax();
8
9
10 }
11
12 int count(int num)
13 {
14     int count=0;
15     while(num!=0)
16     {
17         num=num/10;
18         count++;
19     }
20
21     return (count-1);
22 }
23
24 int rev_dig(int num, int p)
25 {
26     int res;
27     int sum=0;

```

sources
Compile Log
Debug
Find Results
Console

```

27     int sum=0;
28     for(int i=p;i>=0;i--)
29     {
30         res=(num%10);
31         for(int j=i;j>=1;j--)
32             res=res*10;
33         sum=sum+res;
34         num=num/10;
35     }
36     return sum;
37 }
38
39 void printmax()
40 {
41     int prod;
42     int max=0;
43     int a;
44     int m;
45     for(int i=100;i<=999;i++)
46     {
47         for(int j=100;j<=999;j++)
48         {
49             prod=i*j;
50             a=count(prod);
51             m=rev dig(prod,a);
52             m=rev_dig(prod,a);
53             if(m==prod)
54             {
55                 if(m>max)
56                     max=m;
57             }
58         }
59     }
60     printf("%d",max);
61 }

```

Problem: 5 | Highly divisible triangular number

```
#include<stdio.h>
```

```
int main(){
```

```
int num = 85;
```

```
int sum = 0;
```

```

int count = 0;
    int j = 0;
    int i = 1;

printf("First 5 digits with over 10 divisors are : \n");
    for(i = 1 ; i <= num ; i++){
        count = 0;
        sum = 0;
        for(j = 0 ; j <= i ; j++){
            sum += j;}
        for(int k = 1 ; k <= j ; k++){
            if(j%k==0){
                count++;
            }
            if(count==10){
                printf("%d : ",sum);
                for(int n = 1; n<sum ; n++){
                    if(sum%n==0){
                        printf("%d ",n);
                    }
                }
                printf("\n");
            }
        }
    }
}

```

```

    }

    }

    }

return 0;

}

```

```

task 05.cpp
1  #include<stdio.h>
2  int main(){
3      int num = 85;
4      int sum = 0;
5      int count = 0;
6      int j = 0;
7      int i = 1;
8
9      printf("First 5 digits with over 10 divisors are : \n");
10     for(i = 1 ; i <= num ; i++){
11         count = 0;
12         sum = 0;
13         for(j = 0 ; j <= i ; j++){
14             sum += j;}
15         for(int k = 1 ; k <= j ; k++){
16             if(j%k==0){
17                 count++;
18                 if(count==10){
19                     printf("%d : ",sum);
20                     for(int n = 1; n<sum ; n++){
21                         if(sum%n==0){
22                             printf("%d ",n);
23                         }
24                     }
25                     printf("\n");
26                 }
27             }
28
29         }
30     }
31     return 0;
32 }
33

```

Problem: 6 | Random Walk

```
#include<stdio.h>

#include<math.h>

int main()

{

    int x1=0;

    int y1=0;

    int d=0,count=0;

    int x2,y2;

    int sum=0;

    for(int st=0; st>=0; st++)

    {

        //int sum=0;

        scanf("%d",&x2);

        scanf("%d",&y2);

        if(x2==x1&&y2==y1)

            break;

        else

        {d=sqrt(pow(x2-x1,2)+pow(y2-y1,2));

            sum=sum+d;

            x1=x2;

            y1=y2;

            count++;}

    }

    printf("\ndistance is %d\n",sum);

    printf("steps =%d\n",count);

    int avg=sum/count;

    printf("average is =%d\n",avg);
```

return 0;

}

task 06.cpp

```
1  #include<stdio.h>
2  #include<math.h>
3  int main()
4  {
5      int x1=0;
6      int y1=0;
7      int d=0,count=0;
8      int x2,y2;
9      int sum=0;
10     for(int st=0; st>=0; st++)
11     {
12         //int sum=0;
13         scanf("%d",&x2);
14         scanf("%d",&y2);
15         if(x2==x1&&y2==y1)
16             break;
17         else
18         {d=sqrt(pow(x2-x1,2)+pow(y2-y1,2));
19          sum=sum+d;
20          x1=x2;
21          y1=y2;
22          count++;}
23     }
24     printf("\ndistance is %d\n",sum);
25     printf("steps =%d\n",count);
26     int avg=sum/count;
27     printf("average is =%d\n",avg);
```

Problem: 7 | Pattern

Pattern

1

```
#include <stdio.h>

int main()
{

    int i, j, rows;

    printf (" Enter a number to define the rows: \n ");

    scanf("%d", &rows);

    printf("\n");

    for (i = 1; i <= rows; ++i)

    {

        for (j = 1; j <= i; ++j)

        {

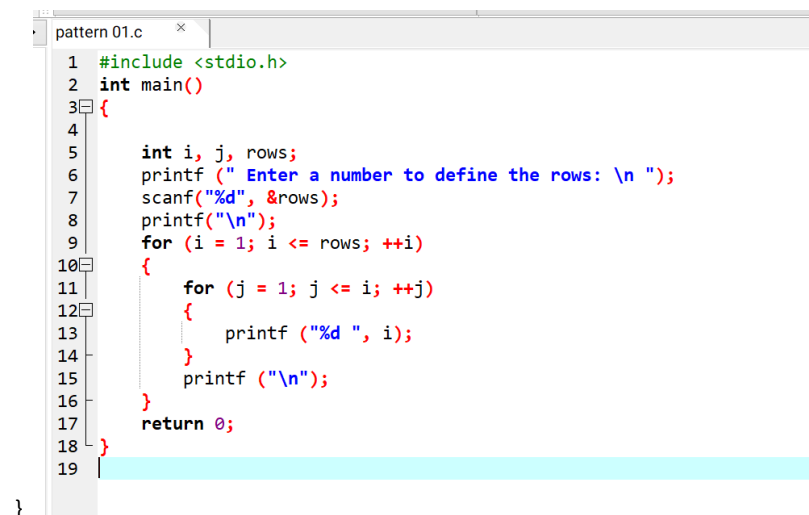
            printf ("%d ", i);

        }

        printf ("\n");

    }

    return 0;
```



The screenshot shows a code editor window titled "pattern 01.c". The code is as follows:

```
1  #include <stdio.h>
2  int main()
3  {
4
5      int i, j, rows;
6      printf (" Enter a number to define the rows: \n ");
7      scanf("%d", &rows);
8      printf("\n");
9      for (i = 1; i <= rows; ++i)
10     {
11         for (j = 1; j <= i; ++j)
12         {
13             printf ("%d ", i);
14         }
15         printf ("\n");
16     }
17     return 0;
18 }
19 }
```

Pattern 2

```
#include <stdio.h>

int main()
{
    int i, j, rows, k;

    printf (" Enter a number to define the rows: \n ");

    scanf("%d", &rows);

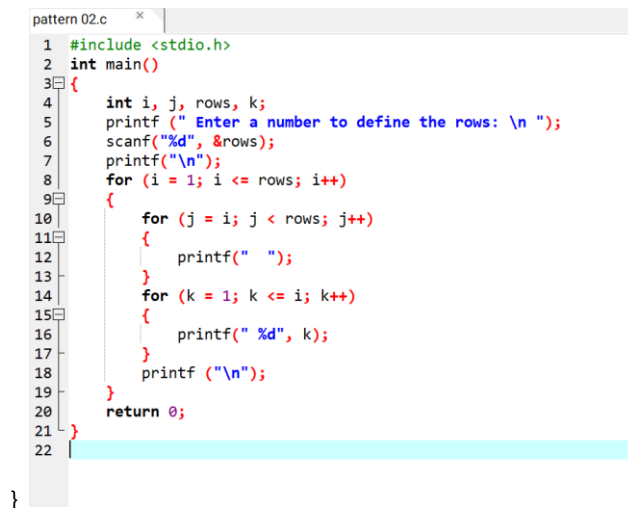
    printf("\n");

    for (i = 1; i <= rows; i++)
    {
        for (j = i; j < rows; j++)
        {
            printf(" ");
        }

        for (k = 1; k <= i; k++)
        {
            printf(" %d", k);
        }

        printf ("\n");
    }

    return 0;
```



```
pattern 02.c
1  #include <stdio.h>
2  int main()
3  {
4      int i, j, rows, k;
5      printf (" Enter a number to define the rows: \n ");
6      scanf("%d", &rows);
7      printf("\n");
8      for (i = 1; i <= rows; i++)
9      {
10         for (j = i; j < rows; j++)
11         {
12             printf(" ");
13         }
14         for (k = 1; k <= i; k++)
15         {
16             printf(" %d", k);
17         }
18         printf ("\n");
19     }
20     return 0;
21 }
22 }
```

Pattern 3

```
#include <stdio.h>

int main()
{
    int i, j, rows;

    printf (" Enter a number to define the rows: \n ");

    scanf("%d", &rows);

    printf("\n");

    for (i = 1; i <= rows; ++i)
    {
        for (j = 1; j <= i; ++j)
        {
            printf ("%c ", 'A' + j - 1);

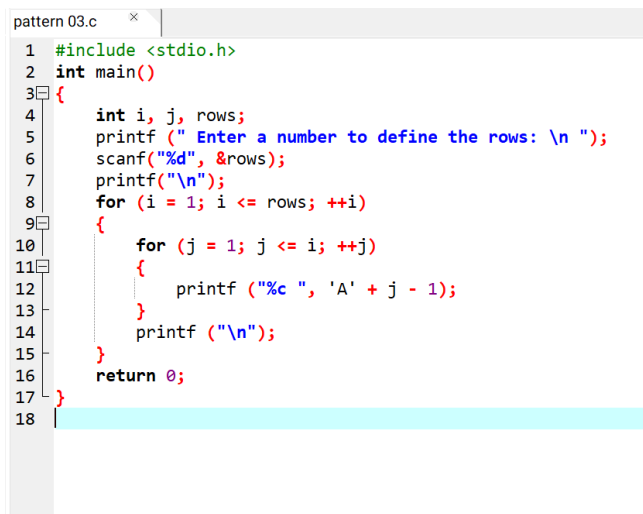
        }

        printf ("\n");

    }

    return 0;

}
```

A screenshot of a code editor window titled 'pattern 03.c'. The editor displays the C program code for Pattern 3, which is identical to the code block above. The code is color-coded: keywords like 'include', 'int', 'for', 'return', and 'scanf' are in blue, string literals are in red, and other identifiers and symbols are in black. Line numbers 1 through 18 are visible on the left side of the editor. The code is as follows:

```
1  #include <stdio.h>
2  int main()
3  {
4      int i, j, rows;
5      printf (" Enter a number to define the rows: \n ");
6      scanf("%d", &rows);
7      printf("\n");
8      for (i = 1; i <= rows; ++i)
9      {
10         for (j = 1; j <= i; ++j)
11         {
12             printf ("%c ", 'A' + j - 1);
13         }
14         printf ("\n");
15     }
16     return 0;
17 }
18
```

Pattern 4

```
#include <stdio.h>

int main()
{
    int i, j, rows, k, m;

    printf (" Enter a number to define the rows: \n");

    scanf ("%d", &rows);

    printf("\n");

    for ( i = 1; i <= rows; i++)
    {
        for ( j = 1; j <= rows-i; j++)
        {
            printf (" ");
        }

        for ( k = 1; k <= i; k++)
        {
            printf ("%d", k);
        }

        for (m = i-1; m >=1; m--)
        {
            printf("%d", m);
        }

        printf ("\n");
    }

    return 0;
}
```

```
pattern 04.c
1  #include <stdio.h>
2  int main()
3  {
4      int i, j, rows, k, m;
5      printf (" Enter a number to define the rows: \n");
6      scanf ("%d", &rows);
7      printf("\n");
8      for ( i = 1; i <= rows; i++)
9      {
10         for ( j = 1; j <= rows-i; j++)
11         {
12             printf (" ");
13         }
14         for ( k = 1; k <= i; k++)
15         {
16             printf ("%d", k);
17         }
18         for (m = i-1; m >=1; m--)
19         {
20             printf("%d", m);
21         }
22         printf ("\n");
23     }
24     return 0;
25 }
26
```

Pattern 5

```
#include<stdio.h>

int main()

{

    int i,j,k;

    for(i=4; i>=0; i--)

        {

            for(k=3; k>=i; k--)

                {

                    printf(" ");

                }

            for(j=i; j>=0; j--)

                {

                    printf("%c ",(i-j)+65);

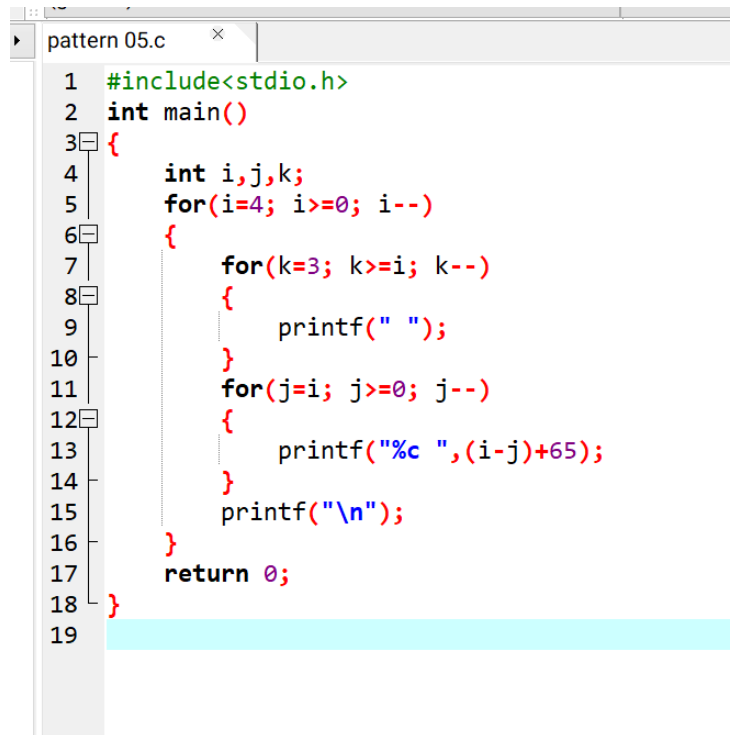
                }

            printf("\n");

        }
```

```
return 0;

}
```

A screenshot of a code editor window titled 'pattern 05.c'. The editor shows a C program with 19 lines of code. The code is as follows:

```
1  #include<stdio.h>
2  int main()
3  {
4      int i,j,k;
5      for(i=4; i>=0; i--)
6      {
7          for(k=3; k>=i; k--)
8          {
9              printf(" ");
10             }
11             for(j=i; j>=0; j--)
12             {
13                 printf("%c ",(i-j)+65);
14             }
15             printf("\n");
16         }
17     return 0;
18 }
19
```

The code is color-coded: keywords like 'int', 'for', 'return', and 'include' are in blue, and string literals like '" "', '"\n"', and '"%c "' are in red. The editor has a light gray background and a vertical line at the end of each line of code.

Pattern 6

```
#include<stdio.h>

int main()
{
    int n=5;
    int px=n;
    int py=n;
    int i,j;
    for(i=1; i<=n; i++)
    {
        for(j=1; j<n*2; j++)
```

```

    {
        if(j>px&& j<py)
        {
            printf(" ");
        }
        else
        {
            printf("*");
        }
    }

    px--;

    py++;

    printf("\n");

}

return 0;

```

```

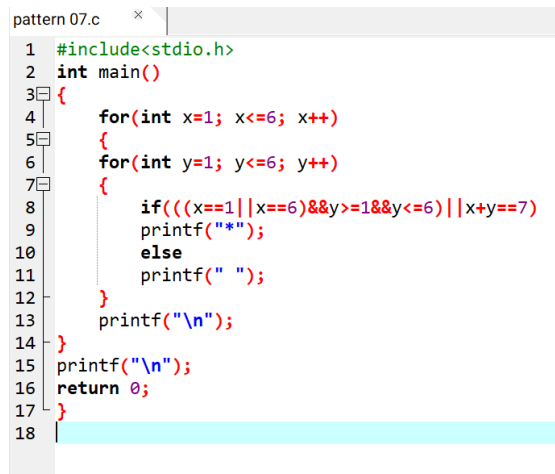
pattern 06.c
1  #include<stdio.h>
2  int main()
3  {
4      int n=5;
5      int px=n;
6      int py=n;
7      int i,j;
8      for(i=1; i<=n; i++)
9      {
10         for(j=1; j<n*2; j++)
11         {
12             if(j>px&& j<py)
13             {
14                 printf(" ");
15             }
16             else
17             {
18                 printf("*");
19             }
20         }
21         px--;
22         py++;
23         printf("\n");
24     }
25     return 0;
26 }
27

```

Pattern 7

```
#include<stdio.h>

int main()
{
    for(int x=1; x<=6; x++)
    {
        for(int y=1; y<=6; y++)
        {
            if(((x==1 || x==6)&& y>=1&& y<=6) || x+y==7)
                printf("*");
            else
                printf(" ");
        }
        printf("\n");
    }
    printf("\n");
    return 0;
}
```



```
pattern 07.c
1  #include<stdio.h>
2  int main()
3  {
4      for(int x=1; x<=6; x++)
5      {
6          for(int y=1; y<=6; y++)
7          {
8              if(((x==1 || x==6)&& y>=1&& y<=6) || x+y==7)
9                  printf("*");
10             else
11                 printf(" ");
12         }
13         printf("\n");
14     }
15     printf("\n");
16     return 0;
17 }
18
```

Pattern 8

```
#include<stdio.h>
```

```
int main()
```

```
{
```

```
for(int x=0; x<5; x++)
```

```
{
```

```
printf("* ");
```

```
}
```

```
printf("\n\n");
```

```
for(int y=0; y<3; y++)
```

```
{
```

```
printf("*      *\n\n");
```

```
}
```

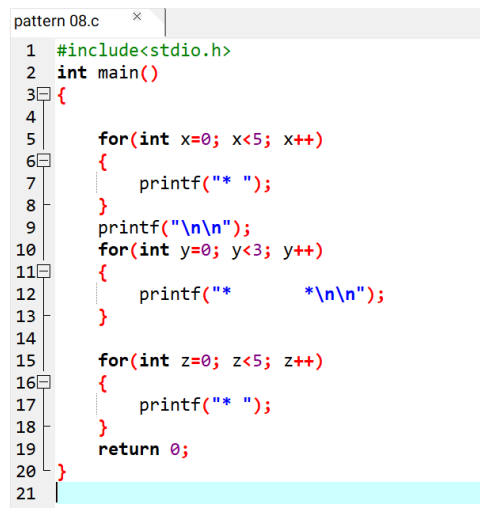
```
for(int z=0; z<5; z++)
```

```
{
```

```
printf("* ");
```

```
}
```

```
return 0;
```



```
pattern 08.c
1  #include<stdio.h>
2  int main()
3  {
4
5      for(int x=0; x<5; x++)
6      {
7          printf("* ");
8      }
9      printf("\n\n");
10     for(int y=0; y<3; y++)
11     {
12         printf("*      *\n\n");
13     }
14
15     for(int z=0; z<5; z++)
16     {
17         printf("* ");
18     }
19     return 0;
20 }
21
```

